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Dear Dr. Catherine Walker:

I am reaching out to ask whether a perspective piece I have been working on with collaborators may be of interest to *Nature Plants*. It is currently entitled ‘The problem and promise of modeling plant budburst under warmer winters.’

The piece focuses on chilling—the concept of cool temperatures required to break dormancy—with a focus on woody plant flowering and leafing, where the concept has major implications for many fruit crops and how well forests can continue to sequester carbon with climate change (1; 2). This is also where forecasts of shifts in chilling with climate change yield almost endless possibilities: current models predict wildly different leafout or flowering times for the same study system and temperature regime (3; 4). Interest in chilling for forecasting the effects of climate change has led to myriad new models, but few insights.

In this perspective we argue that major advances through molecular approaches (e.g., 5; 6; 7; 8) will be slow without a reckoning with the concept of chilling. We highlight fundamental problems with the concept, including the reality that it is an unobserved process, representing the transition between two latent states. The result of this in models of chilling today is hidden assumptions and obscured work-arounds leading to highly uncertain and variable forecasts. We show how new molecular advances may rule out certain common models, and suggest a path towards more mechanistic models that build on and help design experimental results. We close by asking whether the concept of ‘chilling’ may soon be unnecessary for scientific progress.

Our perspective aims to bridge across the often siloed fields that work on chilling—molecular biologists, forest tree and crop modelers, and those focused on forecasting with climate change—to uncover the problems that may limit the potential for rapid progress through new experimental work. As such we leverage existing data from different fields to show how common some of the unspoken problems with current approaches are. I include below a potential abstract and several potential figures.

If you think this article could be of interest to readers of *Nature Plants* or have additional questions, please let me know.

Sincerely,

A handwritten signature in black ink, reading 'Elizabeth M. Wolkovich'.

Elizabeth M Wolkovich
Professor of Forest & Conservation Sciences

Abstract: How warmer winters affect plant budburst has major implications for carbon storage and ecosystem stability. Recent evidence that the advance of spring leafout has slowed with anthropogenic climate change has reignited a decades-old debate over winter chilling—the concept that woody temperate plants require sufficient winter cool temperatures to budburst each spring. The problem with chilling is that it does not correspond with an observable event. This makes models of it rarely falsifiable and of limited use for advancing scientific understanding. In this paper, we review how current research built on the concept of chilling combined with limited data has led to the proliferation of untestable models that are consistent with a wide range of forecasts and provide little insight into actual plant biology. We argue that the concept of chilling should be reevaluated like past theoretical constructs of limited scientific value (such as ‘dormin’ or ‘anthesin’) to help accelerate our understanding of plant development. This can be done by leveraging new insights on the biochemistry and physiology of dormancy to build better models that fit to both experimental and observational data, and by designing better experiments to more rigorously test theory.

- [1] Keenan, T. F. *et al.* Net carbon uptake has increased through warming-induced changes in temperate forest phenology. *Nature Climate Change* **4**, 598–604 (2014).
- [2] Pörtner, H. O. *et al.* *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2022).
- [3] Basler, D. Evaluating phenological models for the prediction of leaf-out dates in six temperate tree species across central europe. *Agricultural and Forest Meteorology* **217**, 10–21 (2016).
- [4] Hufkens, K., Basler, D., Milliman, T., Melaas, E. K. & Richardson, A. D. An integrated phenology modelling framework in r. *Methods in Ecology and Evolution* **9**, 1276–1285 (2018).
- [5] Zhu, P., Lister, C. & Dean, C. Cold-induced arabidopsis frigida nuclear condensates for flc repression. *Nature* **599**, 657–661 (2021).
- [6] Azeez, A. *et al.* Early bud-break 1 and early bud-break 3 control resumption of poplar growth after winter dormancy. *Nature communications* **12**, 1123 (2021).
- [7] Pan, W. *et al.* Epigenetic silencing of callose synthase by VIL1 promotes bud-growth transition in lily bulbs. *Nature Plants* **9**, 1451–1467 (2023).
- [8] Cai, F. *et al.* Molecular regulation of bud dormancy in perennial plants. *Plant Growth Regulation* **102**, 1–11 (2024).

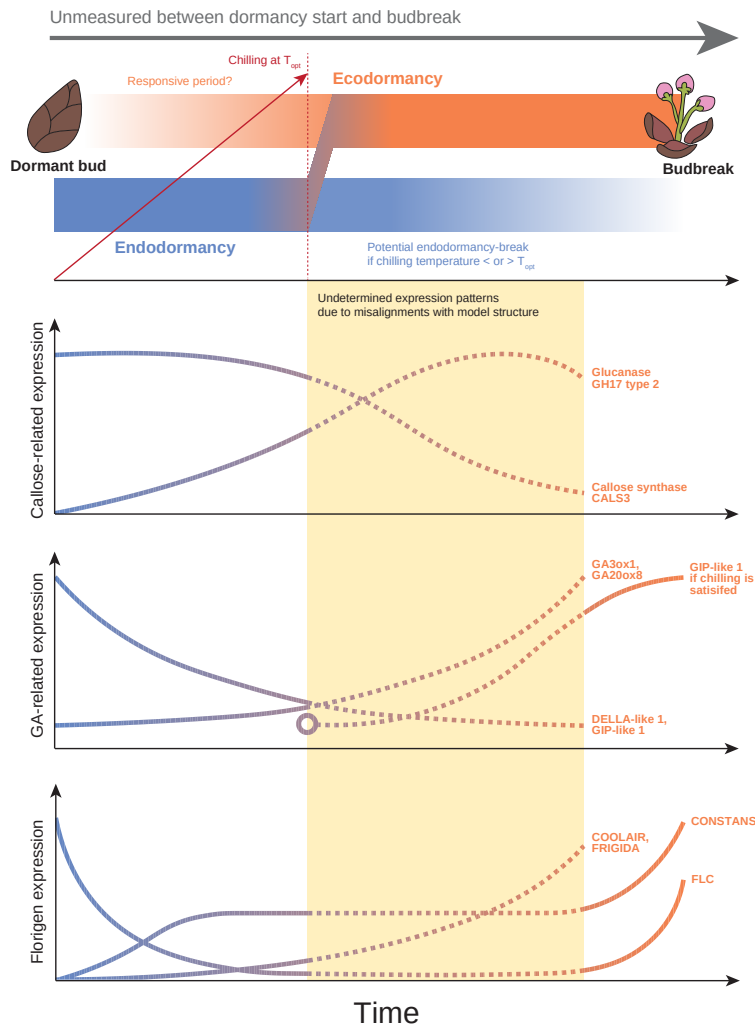


Figure 1: Aligning phenological models and molecular findings. Current phenological models of budbreak (top panel) assume two underlying states that lead to the observed process of budbreak: (1) endodormancy during which plants accumulate sufficient chilling to break endodormancy and transition to (2) ecodormancy, a period during which plants accumulate sufficient ‘forcing’ (warm temperatures) to break bud (flower or leaf). Over 30 variants of this model exist, including those where the phases are sequential (darker shading) or occur in parallel (dark and light shading). Molecular work suggests callose (second panel), alongside gibberellic acid (GA, third panel) and circadian clock-related genes (fourth panel) may all underlie these hypothesized phases.

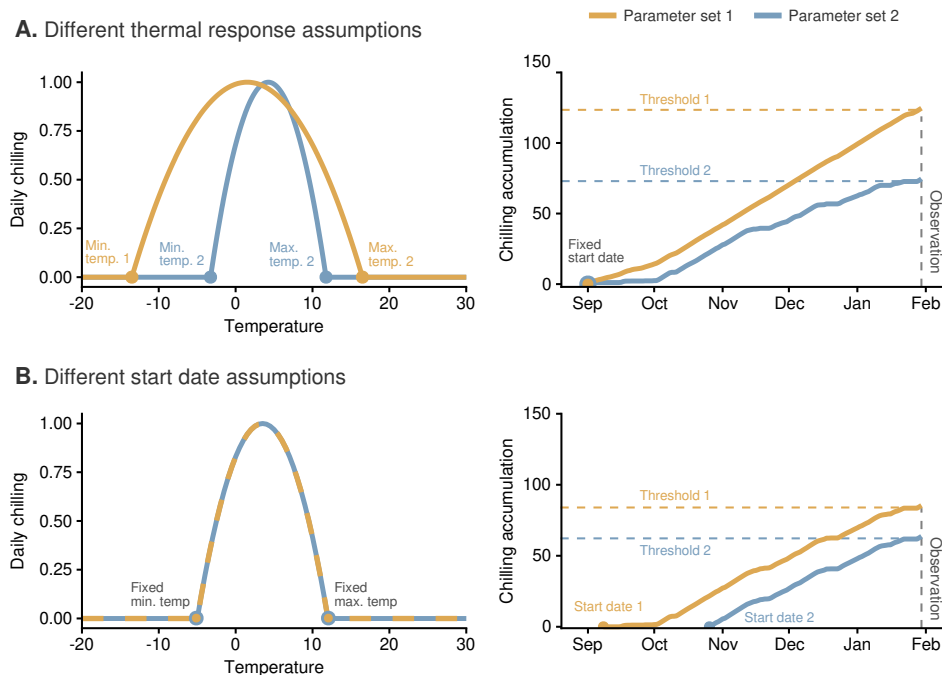


Figure 2: Current models of chilling cannot uniquely estimate the most important aspects of chilling. We illustrate this using a simple chilling model with a symmetrical thermal response curve, assuming the end of the chilling phase (often called endodormancy break) is observed (dashed vertical lines in right panes). In both cases, multiple parameter sets reproduce the same observation equally well. **(A)** Estimating the thermal response curve parameters (min. and max. temperatures) and the chilling threshold jointly (with a fixed start date) leads to many potential solutions. For example, a wider temperature range requires a higher threshold to reach the observed date (yellow), while a narrower range requires a lower threshold (blue). **(B)** The same issue happens when the response curve is fixed but the start date and threshold are estimated jointly. An earlier start date with a higher threshold (orange) leads to the same observed date as a later start date with a lower threshold (blue). These two simple examples illustrate that the parameter space contains multiple indistinguishable solutions, even when the end of the chilling phase is known (which is rare, more often models are built with only data on the end of the next phase—observed via budbreak). This issue would be even worse if the thermal response curve, the threshold, and the start date were all estimated jointly—and when the end of the phase is not directly observed. This reality is present in every model of chilling, but rarely discussed and often not even mentioned.